

Shivang Chopra

Ph.D. in Computer Science at Georgia Institute of Technology

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RESEARCH INTERESTS

Robust Representation Learning: Investigating the algorithmic and geometric principles underlying generalization and efficiency in multimodal foundation models, with a focus on transformer architecture design, and representation structure.

Adaptive Computation and Efficient Inference: Developing methods to scale virtual depth and adaptive computation at inference time, enabling pretrained LLMs to achieve deeper reasoning capabilities without increasing total parameter counts.

Data-Efficient Learning and Training Optimization: Designing data selection, active learning, and manifold-aware training methods to improve sample efficiency, training stability, and generalization in large-scale multimodal and language models.

EDUCATION

Georgia Institute of Technology

Jan 2025 - May 2029 (expected)

Ph.D. in Computer Science

Advisor: Prof. Zsolt Kira [🔗](#)

Publications: CVPR 2026 [2], CVPR(W) 2025 [8], ICLR 2025 [4]

Georgia Institute of Technology

Aug 2022 - Dec 2024

Master of Science in Computer Science

CGPA: 3.9 / 4.0

Thesis: CRAFT: Curriculum Rank Adversarial Fine-Tuning for Robust Vision Language Models [📄](#)

Publications: WACV 2025 [3], CoRL 2023 ??, ICML(W) 2022 [10]

Delhi Technological University

Aug 2016 - May 2020

Bachelor of Technology in Computer Engineering

CGPA: 8.6 / 10.0

Publications: AAAI 2020 [6], IEEE CDS [5]

RESEARCH EXPERIENCE

Sony R&D Labs

Zurich, Switzerland

Computer Vision Research Intern

Sep 2023 - Apr 2024

Mentor: Valentina Cavinato [🔗](#)

Amazon Science

Washington, USA

Applied Scientist Intern

May 2023 - Aug 2023

Mentor: Dr. Tarang Chugh [🔗](#)

Microsoft Research Lab India

Bangalore, India

Research Intern

May 2022 - Aug 2022

Mentor: Dr. Venkat Padmanabhan [🔗](#)

IBM Research Lab

Delhi, India

Research Intern

June 2020 - Aug 2020

Mentor: Dr. Sameep Mehta [🔗](#) , Nistha Madaan [🔗](#)

PUBLICATIONS

In Progress

- [1] **Thinking Deeper Without Growing Larger: Scaling Virtual Depth in Pretrained LLMs for Enhanced Reasoning**

Shivang Chopra, Zhang Xi-Jia, Karmesh Yadav, Yusuf Ali, Chengyue Huang, Zsolt Kira.

Conferences and Journals

- [2] **The Geometry of Robustness: Optimizing Loss Landscape Curvature and Feature Manifold Alignment for Robust Finetuning of Vision-Language Models**

Shivang Chopra, Shaunak Halbe, Chengyue Huang, Brisa Maneechotesuwan, Zsolt Kira.

Paper accepted at IEEE/CVF Conference on computer vision and pattern recognition, [CVPR 2026](#) [📄](#)

- [3] **Refining Text-to-Image Generation: Towards Accurate Training-Free Glyph-Enhanced Image Generation**

Sanyam Lakhanpal, Shivang Chopra, Vinija Jain, Aman Chadha, Man Luo

Published at IEEE/CVF Winter Conference on Applications of Computer Vision, [WACV 2025](#) [📄](#)

- [4] **Directional Gradient Projection for Robust Fine-tuning of Foundation Models**

Chengyue Huang, Junjiao Tian, Brisa Maneechotesuwan, Shivang Chopra, Zsolt Kira

Published at the International Conference on Learning Representations, [ICLR 2025](#) [📄](#)

- [5] **Attention-Net: An Ensemble Sketch Recognition Approach using Vector Image**

Shivang Chopra*, Gaurav Jain*, Suransh Chopra*, Anil Singh Parihar

Published in IEEE Transactions on Cognitive and Developmental Systems, [IEEE CDS 2022](#) [📄](#)

- [6] **Hindi-English Hate Speech Detection: Author Profiling, Debiasing, and Practical Perspectives**

Shivang Chopra, Ramit Sawhney, Puneet Mathur, Rajiv Ratn Shah

Published at the AAAI Conference on Artificial Intelligence, [AAAI 2020](#) [📄](#)

Workshops/Preprint

[8] **MedMoE: Modality-Specialized Mixture of Experts for Medical Vision-Language Understanding**
Shivang Chopra, Lingchao Mao, Gabriela Sanchez-Rodriguez, Andrew J. Feola, Jing Li, Zsolt Kira
Published at the [MMFF-BIOMED workshop at CVPR 2025](#). 

[9] **Source-free domain adaptation with diffusion-guided source data generation**
Shivang Chopra, Suraj Kothawade, Houda Aynaou, Aman Chadha. arXiv 2024 

[10] **Active Data Discovery: Mining Unknown Data using Submodular Information Measures**
Suraj Kothawade, **Shivang Chopra**, Saikat Ghosh, Rishabh Iyer
Published at the [RealML workshop at ICML 2022](#). 

Patent

[P1] Hybrid Environment for Interactions Between Virtual and Physical Users (**US 18/358,485**), 2024

AWARDS AND ACHIEVEMENTS

Graduate Student Scholarship , Georgia Institute of Technology	2023,2024
Research Excellence Award , Delhi Technological University	2020
Student Research Travel Grant , Microsoft Research India	2020
Qualified for Onsite Regionals , ACM ICPC (International Collegiate Programming Contest)	2018

RELEVANT COURSEWORK

Ph.D. / Master Of Science: Machine Learning, Vision Language Models, Deep Learning for Robotics, Game AI, Natural Language Processing, Learning and Memory, Cognitive Psychology
Bachelor of Technology: Machine Learning, Artificial Intelligence, Natural Language Processing, Distributed Systems, Operating Systems, Computer Networks, Information and Network Security, Computer Organization and Architecture

TECHNICAL SKILLS

Programming Languages	Python, Bash, C, C++
Deep Learning Frameworks	PyTorch, TensorFlow, JAX, Accelerate, Pytorch Lightning
Distributed System Tools	Slurm, Docker, Kubernetes

COMMUNITY AND ACADEMIC SERVICES

Reviewer: CVPR 2026, NeurIps 2024-25, ICLR 2025, ICML 2025
Student Volunteer: AAAI 2020, ICRA 2025
Project Lead, AI@GT (Georgia Tech): Mentoring a group of three undergraduate students to build self-evolving embodied agents that learn by interacting with their environment.
Admin, GT Skynet: Part of the team maintaining Georgia Tech’s SLURM-based GPU cluster to support ML research and coursework.